

MECHA-MIX: THE ALL-IN-ONE MIXED DRINK PREPARATION MACHINE

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ABSTRACT

Mecha-Mix is an automated mixologist machine designed to precisely dispense, and mix to provide multi liquid ingredient beverages with strict sanitation protocols following the FDA regulations. The project was developed to address the growing demand for automated manual tasks, more specifically the need for high-precision, hygienic, and beverage preparation. The design of the Mecha-Mix was developed from a comprehensive analysis of prior beverage machines, evaluating the general positives of commercial patents and fluidic technologies. The Mecha-Mix functions were split into critical subsystems including refrigeration, mixing, and automated cleaning cycles, where each of the subsystems have rigorous target specifications such as but not limited to volumetric accuracy of ± 0.1 mL and preparation times under 60 seconds.

In order to achieve these, the main mechanism relies on high-precision peristaltic pumps to deliver controlled, contamination-free flow through isolated tubing pathways. The machine has modular, clean-in-place design to further comply with NSF/ANSI and FDA food standards in order to allow rapid disassembly, flushing, and maintenance of all in contact fluid components. All these design decisions create a cohesive system that prioritizes consumer safety and consistent quality. The final prototype passed all eight performance tests and was manufactured at a total component cost of \$834.

NOMENCLATURE

Organizational Abbreviations

FDA	Food and Drug Administration
NSF	National Sanitation Foundation
ANSI	American National Standards Institute
CFR	Code of Federal Regulations
UL	Underwriter Laboratories
IEC	International Electrotechnical Commission

Technical Abbreviations

TEC	Thermoelectric Cooler
CIP	Clean-In-Place
PWM	Pulse-Width Modulation

NC	Normally Closed
MCU	Microcontroller Unit
PSU	Power Supply Unit
PLA	Polylactic Acid
COP	Coefficient of Performance
FEA	Finite Element Analysis
CFD	Computational Fluid Dynamics
BoM	Bill of Materials
RPM	Revolutions Per Minute

1. INTRODUCTION

1.1 Problem Definition

Mecha-Mix is motivated by an inefficiency gap within the commercial and consumer beverage sectors. Traditional drink mixing, whether performed manually by staff or executed by basic dispensing machines, is prone to inconsistency, is time-consuming, and is heavily dependent on skilled labor, leading to uneven taste and portion sizes. Furthermore, high-traffic venues struggle with long wait times, stressed staff, and sanitation concerns, problems exacerbated by the large operational footprints of traditional mixing stations. The project's core motivation, inspired by models of rapid-service efficiency like Blank Street Coffee, is to dramatically reduce service times and the required physical footprint using technology to improve the beverage making process, allowing customers to promptly receive a perfectly mixed beverage with minimal operational strain on the business, and allow the people to be people.

The significance of the Mecha-Mix project lies in its ability to simultaneously solve these operational problems while delivering a premium customer experience. The uniqueness of the design is captured in the value proposition of delivering "Perfect Taste, Every Time" with unparalleled precision achieved through ultra-high volumetric accuracy while operating complex motion within a compact, cost-constrained, and food-safe enclosure.

To address these challenges, industrial principles were applied and specific technological assumptions were made. The solution relies on a strong mechanical foundation, incorporating principles from Fluid Mechanics, Kinematics, Machine Design,

and Controls. Assuming the use of a positive displacement pump system to guarantee volume independent of fluid viscosity is essential to meeting the accuracy goal. Temperature control, to make sure that drinks are delivered cold to the customer, is also crucial to the solution with multiple options explored. Finally, the integration of an automated mixing process to ensure final homogenization, completing the perfect taste profile. The entire system is also designed for hygiene, employing a modular chassis and a closed-loop cleaning system to align with food safety standards.

Thus to solve all these challenges and incorporate the mixing, dispensing, cooling, and cleaning processes effectively the machine should be able to feed in a minimum of three ingredients cooled to a safe and enjoyable temperature, have commercial level accuracy, mix them into a single homogenous beverage in an easy-to-use countertop machine under the provided \$1000 budget while making minimal performance trade-offs.

1.2 Prior Art

Prior to the idea of Mecha-Mix, there were other products on the market that served a similar purpose. From reviewing various patents and other products, the sanitary was found to be of utmost priority, and we achieve this measure by complying to food equipment standards set by NSF, ANSI, and other organizations. Another finding was the potential tradeoffs based on the choice of pump. A peristaltic pump offers more sanitary benefits, while syringe pumps provide dispensing of higher accuracy with liquids of different viscosities. Gantry systems are used for the project's kinetic element. They are known for their robustness and accuracy, being a good alternative to robotic elements. This research puts this project on the right path on the specific metrics and elements of Mecha-Mix.

The following sections separate the market into two distinct segments. The Consumer and Luxury segment which are products targeting home users and enthusiasts and the Commercial and Industrial segment which are designed for high-volume venue use. The sections highlight the target demographics, the dispensing mechanism, the sanitation protocols and the overall system.

1.2.1 Consumer and Luxury Segments.

Bartesian targets the mainstream consumer segment and prioritizes ease of use over mixology complexity. Utilizing a capsule-based pump dispensing mechanism, Bartesian identifies the cocktail via a barcode on the inserted capsule, which contains all the necessary bitters, juices, and concentrates [1]. It then activates specific pumps to draw the correct raw spirit from attached reservoirs, mixing it with the capsule contents as it dispenses. This process eliminates the need for the user to measure ingredients or stock perishable mixers, offering unmatched convenience.

While sanitation features are not heavily advertised, the design inherently minimizes mess because the sticky ingredients are contained within the disposable capsule. The lack of specified sanitation cycles means users must manually ensure the water reservoir and needle remain clean. Ultimately, the system is a static appliance with no kinetic or moving show parts, designed to sit quietly on a countertop. Its capsule system remains both its greatest strength and weakness: while it streamlines the process,

it is highly restrictive, preventing users from using their own fresh juices or creating custom recipes, thus limiting them to the flavors provided by the manufacturer.

Next, Barsys 360 is aimed at the luxury consumer segment and differentiates itself by utilizing a unique scale-based dispensing mechanism rather than traditional pumps [2]. Its accuracy is determined by real-time weight measurement. The manufacturer has not explicitly detailed the sanitation features or internal cleaning protocols for the tubing.

The device features a moving dispense head as its primary kinetic element, which is intended to add a futuristic, architectural visual flair to a home bar. Unfortunately, user reports indicate that this moving component can be messy and clunky, failing to deliver the seamless experience promised by its design. Additionally, the system suffers from potentially inconsistent accuracy.

1.2.2 Commercial and Industrial Segments.

Makr Shkr is arguably the most famous robotic bartender system in the world, frequently found as a centerpiece on cruise ships and in high-end Las Vegas venues. Targeting the commercial and event segment, it utilizes a fully articulated robotic arm dispensing mechanism to physically manipulate bar tools [3]. Unlike simple pump-boxes, this system is a marvel of industrial automation that can grab cups, muddle ingredients, shake cocktail shakers, and the drinks with a high degree of flexibility.

The core value proposition is its kinetic element: a fully choreographed robotic motion that mimics the gestures of a human bartender to create a spectator event that draws crowds and encourages social media sharing. While sanitation features are unclear, the open nature of the system implies a need for external cleaning protocols to ensure the work area remains hygienic. Despite its visual appeal, Makr Shkr is defined by its large scale, high complexity, and extreme expense, making it less of a dispenser and more of an attraction.

Similarly, ADAM positions itself in the high-end commercial segment by employing a sophisticated dual-arm robotic dispensing mechanism designed to mimic human bartenders [4]. While there are no specific public claims regarding its pouring accuracy, it places a heavy emphasis on safety and hygiene, boasting sanitation features that are NSF certified and require approximately 20 minutes of daily maintenance.

The system's primary value proposition is its kinetic element, characterized by a complex two-armed robotic motion that creates a theatrical, highly shareable experience for venue guests. However, this spectacle comes with significant logistical downsides, including a very high financial entry point and a massive physical footprint. Consequently, venue owners often face difficult challenges regarding its complex integration into standard bar layouts and workflows.

Sidework is also strictly designed for the commercial segment, relying on a high-efficiency pump dispensing mechanism to handle volume. Its precision is a primary selling point, with accuracy claimed to be within 1/20th of an ounce to significantly reduce liquor variance and waste [5]. To support high-turnover environments, the system's sanitation features include an automatic self-cleaning function, which minimizes the manual labor required by staff during closing duties.

However, the unit is purely functional and has no kinetic elements, meaning it operates as a background utility rather than a visual attraction for customers. Furthermore, its public technical specifications are currently limited, which can make it difficult for engineers to assess its long-term maintenance needs.

Mixologiq targets the commercial sector, utilizing a standard pump dispensing mechanism to serve mixed drinks with speed and consistency [6]. Although specific data regarding its pouring accuracy appears to be incomplete or cut off in documentation, the system emphasizes hygiene with automatic cleaning cycles and parts that are easily accessible for service.

It incorporates a semi-automatic kinetic element to assist in the drink-making process, bridging the gap between manual and fully automated service. However, because it is enclosed and utilitarian, it is significantly less visually engaging for customers compared to systems with articulated robotic motion. This makes it more of a productivity tool for service wells rather than a centerpiece for a front-of-house bar experience.

Bevi targets the commercial/office segment, utilizing a pump dispensing mechanism and flavor pods [7]. There are no claims about their accuracy, but the sanitation features are NSF certified with touchless dispensing and an advanced filtration system. Bevi has no kinetic elements and is primarily used for water and flavored water instead of mixed beverages.

1.2.3 System Comparison Summary.

The following table summarizes the key operational parameters of the systems analyzed above.

TABLE 1: Comparison of Automated Beverage Dispensing Systems

System	Target Market	Mechanism	Kinetic Elements	Key Characteristics
Bartesian	Mainstream Consumer	Capsule-based Pump	Static	Convenience focus; restrictive capsule ecosystem; manual cleaning.
Barsys 360	Luxury Consumer	Scale-based (Weight)	Moving Dispense Head	Futuristic design; messy visuals; inconsistent accuracy.
Makr Shagr	Commercial / Events	Articulated Robotic Arm	High (Choreographed)	Spectator attraction; high complexity and cost; industrial scale.
ADAM	High-end Commercial	Dual-arm Robotic	High (Theatrical)	NSF certified hygiene; massive footprint; difficult integration.
Sidework	Commercial (Volume)	High-efficiency Pump	Static	High precision (1/20 oz); auto self-cleaning; purely functional.
Mixologiq	Commercial	Standard Pump	Semi-Automatic	Bridging tool; accessible service parts; enclosed design.
Bevi	Office / Commercial	Pump & Flavor Pods	Static	NSF certified; touchless; focus on flavored water.

2. REQUIREMENTS

The final design and construction of Mecha-Mix is based on a precise set of technical specifications derived directly from core customer needs for quality, efficiency, and hygiene. These specifications led to measurable engineering metrics that define

the success of the prototype, bridging the gap between customer expectation and successful mechanical execution.

2.1 Customer Needs and Core Goals

The project’s foundation addresses three primary shortcomings in existing automated beverage dispensers, which guide all subsequent engineering choices. First, there is a critical need for precision and consistency because traditional gravity-fed or time-based systems yield inconsistent results due to varying liquid viscosities. Mecha-Mix must solve this to guarantee "Perfect Taste, Every Time." There is also a demand for improved operational efficiency and hygiene, addressing high labor costs and significant sanitation concerns in commercial environments. By satisfying these needs, the design delivers an enclosed, sanitary system.

These concerns are backed by data from various sources. According to Toast, 29% of full-service restaurants are short on bartenders, making it the most in-demand role [8], while a CDC study showed that there were cross-contamination actions at 63% of the restaurant establishments they surveyed [9]. Mecha-Mix directly addresses both of the issues through automated dispensing that completely eliminates the need for labour.

2.2 Functional and Performance Specifications

The highest priority functional requirement is to distribute liquid ingredients with a volumetric accuracy of 0.1 ml for all volumes distributed, regardless of viscosity or temperature. This high threshold necessitates the use of a positive displacement pump as the primary metering device, as these pumps physically push a fixed volume of liquid, ensuring volume is constant. The system must be capable of mixing and dispensing a full range of serving sizes, from a small 45 ml cocktail volume to a large 250 ml serving size per cycle. The final complete beverage preparation cycle time is constrained to 60 seconds per drink (from selection to delivery), a threshold that balances commercial speed requirements with the time necessary for the deliberate, choreographed sequence.

2.3 Thermal, Mixing, and Flow Control

To ensure ingredient quality, the refrigerated storage must hold all perishable liquids within a tight range of $5 \pm 2^{\circ}\text{C}$. This temperature stability is achieved using a compact thermoelectric (Peltier) cooling module. Furthermore, the system must include an automated final homogenization action immediately after dispensing to fully blend ingredients of differing densities. This mixing is achieved using a mechanical stirring rod mechanism. To ensure precise flow control during both dispensing and cleaning, food-grade solenoid valves are integrated into the fluid pathway to create sharp cutoffs and prevent unwanted liquid entry or backflow. Additionally, the system must accept and execute user-customized recipes and ratio adjustments via a touchscreen to satisfy both personalization and hygiene goals.

2.4 Hygiene and Serviceability Specifications

Hygiene is managed by rigorous, measurable serviceability standards. The automated quick cleaning cycle utilizes a Clean-In-Place (CIP) system running water and sanitizer, and must be

completed within 2 minutes to minimize operational downtime and prevent microbial growth in the fluid lines. For comprehensive maintenance, the deep cleaning process requires that all fluid contact components (reservoirs, pumps, tubing) be fully disassembled and reassembled within 15 minutes. This mandates a modular design with quick-disconnects for streamlined maintenance. Furthermore, serviceability for routine refills is simplified by requiring that all refill access points be consolidated to one primary service location/side, accessible from a standing position. To define a measurable cleanliness threshold, the automated quick-clean cycle is deemed successful when clear water dispensed immediately after the cycle shows no visible bleed relative to a clear control sample. This criterion provides an objective, repeatable pass/fail benchmark for cross-contamination prevention.

2.5 Codes, Standards, and Physical Constraints

The final design is bound by both industry codes and prototype fabrication constraints:

- **Sanitation Standards:** The product must be designed to comply with NSF/ANSI Food Equipment requirements (specifically referencing NSF/ANSI 18) [10]. Additionally, all fluid-contact materials must comply with FDA 21 CFR 174 and 177 (Polymers) to prevent toxic leaching and guarantee public safety [11].
- **Structural Integrity:** The structural design must be safe and rigid, prioritizing a frame capable of supporting maximum loads and preventing vibration or deflection that would compromise dispensing accuracy. The system's low-voltage electrical components must comply with UL 1082 / IEC 60335-1 guidelines.
- **Physical Envelope:** The device must meet strict dimensional requirements for compact design: $0.50 \times 0.35 \times 0.50 \text{ m}^3$, ensuring it fits under standard countertops with a maximum mass of 12 kg.
- **Manufacturability:** The design is constrained by a maximum allowable cost of \$1,000 for materials and must be manufacturable within the MakerSpace using 3D printing, laser cutting, and machining. Note: Welding is not allowed under the design constraints for Senior Design.

3. PRODUCT DESIGN

The concept development process for Mecha-Mix transformed the high-level customer requirements defined in the previous section into a concrete engineering architecture. This process began with a rigorous problem decomposition, illustrated in the Functional Diagram (Figure 1), which served to isolate the system's critical inputs, such as liquid ingredients, user signals, and electrical energy, and map them to the necessary internal processes. By treating the machine as a system of "black box" functions, where specific subsystems required for operation: accepting and refrigerating liquids, measuring precise dispensing amounts, mixing ingredients to homogeneity, and executing cleaning cycles for sanitation. This diagram highlights the flow

TABLE 2: Summary of Target Specifications

Parameter	Threshold
Volumetric accuracy	$\pm 0.1 \text{ mL}$
Ingredient channels	≥ 5
Dispense volume range	45–500 mL
Preparation cycle time	$< 60 \text{ s}$
Cooling temperature	$5 \pm 2^\circ \text{C}$
Quick clean cycle time	$< 2 \text{ min}$
Deep clean time	$< 15 \text{ min}$
Valve leak rate (static)	0 drops/min
Cleaning residue	Visually clear vs. control sample
Physical envelope	$\leq 0.50 \times 0.35 \times 0.50 \text{ m}^3$
Total component cost	$\leq \$1,000$

from physical and signal inputs to the final outputs, which include the mixed drink, clean valves, and waste liquids.

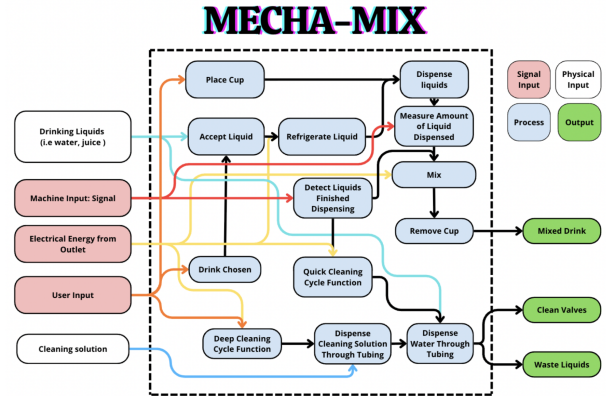


FIGURE 1: Functional Diagram illustrating the system inputs, internal processes (storage, metering, mixing, cleaning), and outputs.

To address the specific functional needs identified in the diagram, we utilized a Morphological Chart (Table 3) to generate a wide range of design alternatives for the three primary subsystems: Cooling, Dispensing, and Mixing. This ideation phase involved evaluating distinct mechanisms to determine the most effective solution for each function. For cooling, we compared standard compressor-based refrigeration against thermoelectric (Peltier) cooling and gravity/passive options. For the critical dispensing function, we weighed the benefits of gravity-fed valves and syringe pumps against positive displacement peristaltic pumps. Similarly, the mixing function was evaluated through options including internal oscillation, magnetic stirring, and internal mechanical stirring.

The sections below go into more detail regarding the rationale behind the chosen concepts highlighted in Table 3.

3.0.1 Cooling Theoretical Background and Concept Development.

The first approach for cooling the Mecha-Mix was the

TABLE 3: Morphological Chart

Function	Option 1	Option 2	Option 3	Chosen
Cooling	Compressor	Thermoelectric	N/A	Thermoelectric
Dispensing	Gravity	Peristaltic	Syringe Pump	Peristaltic
Mixing	Internal	Oscillation	Magnetic Stir	Internal
Cleaning	Manual	Disposable Lines	CIP	CIP
UI	Buttons	App Control	Touchscreen	Touchscreen
Reservoir Access	Multi-side	Rear Access	Single-side Service	Rear Access

traditional refrigeration system. This method had several advantages, such as the ability to reach large temperature differentials, a track record of use in commercial refrigeration, and wide usage in cooling food and beverages. However, it does have some drawbacks; with a system as small as the Mecha-Mix, a whole refrigeration system would be excessive. The refrigeration system requires a compressor, refrigerants, and additional mechanical components. This would not only increase the weight of the system but also create more noise, making it unsuitable for a small form factor household use machine.

The thermoelectric cooler (TEC) was selected for its reliability, size, and cooling performance. Specifically, we implemented a dual-module Peltier assembly that functions as a solid-state heat pump. This mechanism allows for direct integration into the chassis, significantly reducing the physical footprint compared to a compressor. The only major drawback of thermoelectric cooling in the temperature range desired for this product is that it has significantly worse power efficiency than traditional refrigeration. Due to the small size and interior volume of the Mecha-Mix, we do not anticipate any significant power draw from the TEC.

The TEC paired with fans on both the hot and cold plates allows it to achieve precise and reliable temperature control. The TEC operates using the Peltier effect, pulling heat away from the drink reservoir while simultaneously rejecting that heat through the hot-side heat sink. By implementing a fan on each side, the system significantly improves heat transfer. The fan on the cold side enhances cooling efficiency by continuously pushing chilled air from the heat sink to the reservoir, while the fan on the hot side prevents thermal buildup and allows the heat to dissipate at a much faster rate. The diagram for which is shown in Figure 2.

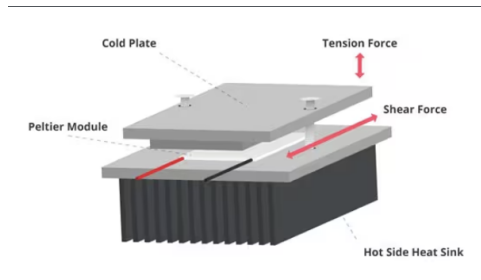


FIGURE 2: Diagram of the thermoelectric cooler (TEC) showing the cold plate, Peltier module, and hot-side heat sink. Adapted from [12].

Compared to compressor-based refrigeration, this dual-fan TEC assembly remains compact, lightweight, and free of refrigerants or moving mechanical components, making it ideal for a small interior volume like that of the Mecha-Mix. In addition, when comparing cost-effectiveness, the thermoelectric cooler outperforms other methods of cooling.

To validate the cooling geometry, we conducted Computational Fluid Dynamics (CFD) simulations to analyze the thermal distribution within the chamber as shown in Figure 3. The integration strategy relies on active circulation, where fans circulate cold air through the internal storage chamber to cool the liquid containers.

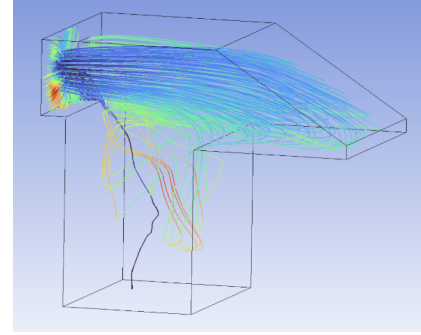


FIGURE 3: CFD simulation results: $T_{min} = 4.5^{\circ}\text{C}$, airflow distribution pathways; Boundary conditions: $Q_{inlet} = 15 \text{ ft}^3/\text{min}$, $\Delta T = 8^{\circ}\text{C}$, $T_{amb} = 25^{\circ}\text{C}$.

The simulation results confirmed the system's viability with the following key metrics:

- **Temperature Targets:** The simulation achieved a minimum temperature of 4.5°C , successfully surpassing the earlier design target of 5°C .
- **Airflow Optimization:** Initial analysis of the "Age of Air" and velocity streamlines indicated that the current geometry allowed some cold air to bypass the lower bottle cavities, potentially reducing convective efficiency.
- **Design Refinement:** To address this bypass issue, the final design will feature vents directed towards the bottles and a minimal cavity size. This modification will force airflow more directly across the entire liquid volume.

Using all of this information the CAD model for the thermoelectric cooling module was represented as shown in Figure 4 below.

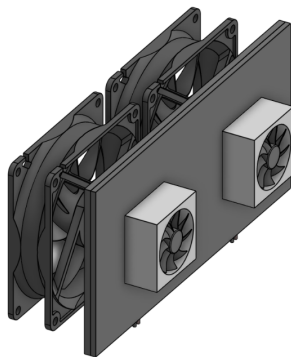


FIGURE 4: Thermoelectric Cooler Design

3.0.2 Dispensing Theoretical Background and Concept Development.

The first dispensing approach was time-based dispensing. This method is fairly simple; all it takes is an open valve and a timer. The way it works is opening a valve for a certain amount of time, allowing fluid to come through; however, it assumes all the liquids have a constant flow rate, which isn't always true. Flow rate is dependent on the viscosity, temperature, level, and pressure of the fluid. In addition, there is no feedback controller, so the system is basically dispensing blindly. With no method of measuring the amount of liquid that comes out, the volume of liquid dispensed could vary. Due to these factors changing with every liquid, the time-based dispensing wouldn't provide consistent results. Therefore, this method of dispensing wasn't chosen.

The volumetric peristaltic pump, a similar low cost version of syringe pumps, was selected as the dispensing mechanism because it provides precise and accurate dosing, which is critical for achieving the ± 0.1 ml accuracy requirement. Unlike time-based or gravity-feed methods, the peristaltic pump uses a rotating set of rollers. These rollers compress the flexible tubing and physically push a known volume of liquid forward with each revolution. This makes the output independent of viscosity, temperature, or bottle fill level, ensuring that syrups, water, and concentrated flavorings all dispense at consistent ratios.

The pump's design, shown in Figure 5, also allows for a more hygienic dispense. Since the fluid only contacts the internal tubing and never the pump body, this reduces the risk of contamination and allows for natural integration with the cleaning cycle. The tubing can be easily replaced or sanitized without disassembling mechanical components.

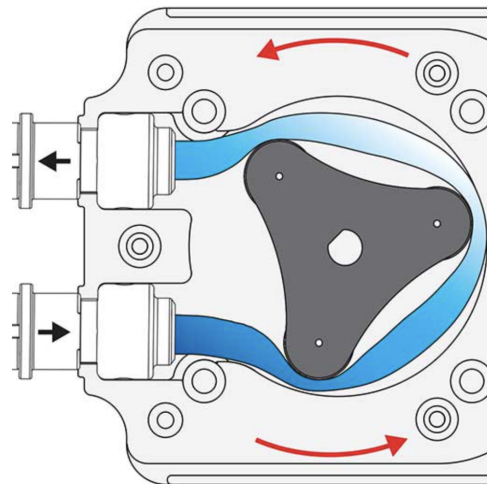


FIGURE 5: Peristaltic pump mechanism: rotating rollers compress flexible tubing to displace a fixed volume of fluid per revolution, ensuring viscosity-independent accuracy. Adapted from [13].

From a systems perspective, the peristaltic pump allows for precise control through motor RPM mapping, allowing us to convert the pump's rotational motion directly into volumetric output. This relationship guarantees consistent flavor profiles and significantly simplifies calibration. Ultimately, the peristaltic pump offered the best balance of accuracy, reliability, hygiene, and integration flexibility, making it the optimal choice for Mecha-Mix's dispensing subsystem.

3.0.3 Mixing Theoretical Background and Concept Development.

The two initial methods of mixing considered were magnetic pill mixing and shaker ball mixing. First, magnetic pill mixing, while this method does offer simplicity and hygiene, it will struggle with fluids of higher viscosity. This would be due to magnetic decoupling; the pills would struggle to mix while the motor runs.

Secondly, the shaker ball, while this generally is a good method of mixing liquids of different viscosity, it is not suitable for the design. This method is highly aggressive and would cause high vibrations, which would not only make the system loud but would also cause strain on the whole machine, leading to potential causes of leaks, parts loosening, and mechanical failure.

As a result, neither of these methods were chosen, as both methods presented limitations that conflicted with the project's goals of reliability, cleanliness, controlled mixing, and mechanical simplicity.

For the mixing subsystem, we selected a metal stirring rod shown above (Figure 6) because it offers the most controllable, reliable, and mechanically robust method for mixing a wide range of beverage ingredients. Unlike the magnetic stirrers or shaker-style systems, the design couples a motor shaft directly to the stirrer, allowing us to deliver high, consistent torque capable of mixing thick syrups and liquids of all sorts of viscosity.

The mixer could be 3D printed using PLA (see Appendix B for rationale), which is safe for food contact, easy to produce, and cost effective but was chosen to be purchased out of stainless

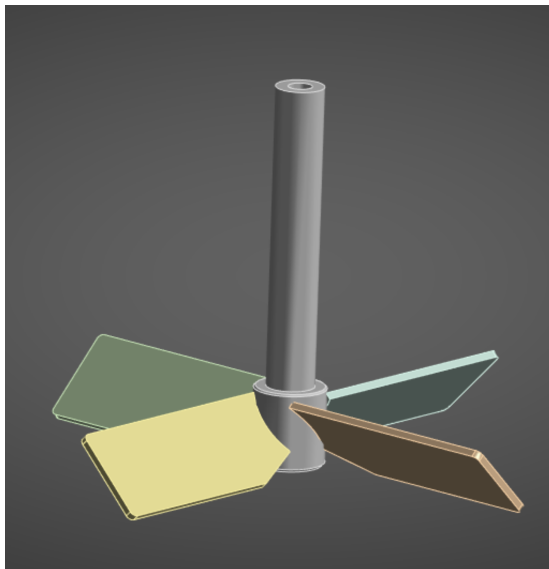


FIGURE 6: 3D Model of Mixer

steel.

Mechanically, the system is also highly stable: the mixing chamber remains stationary while only the stirrer rotates, eliminating the vibration, sealing challenges, and structural stresses associated with shaker-style mechanisms. This stability improves system longevity and simplifies sealing, as no dynamic shaking forces are transmitted to the walls.

Overall, the impeller provides the best balance of power, reliability, and simplicity, making it the optimal solution for delivering consistent drink quality in Mecha-Mix.

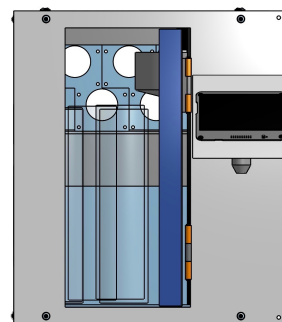
3.0.4 Cleaning Theoretical Background and Concept Development. A flush-through cleaning cycle was selected as the dedicated cleaning method for Mecha-Mix. There would be 2 different types of cleaning consisting of a quick cycle and deep cleaning. The quick cycle follows a run through the tubing with water and flush it out with water. This is in order to clean the tubing between the runs of dispensing different liquids. The second is the deep cycle which is for the end of day, which is more towards the critical cleaning aspect. It adopts a clean in place system which the system runs with water, sanitizer, and air through all fluidic paths. The way this cycle works is by running a cleaning solution and then a fresh water rinse through the system's tubing. This approach leverages the existing tubing network and peristaltic pump, allowing us to thoroughly sanitize the system without adding additional hardware. By circulating the cleaning solution through every fluid pathway, including the pump tubing, mixing chamber inlet, and dispensing line, we ensure a clean path.

Compared to mechanical scrubbing mechanisms or multi-line self cleaning systems, the approach offers a low-cost, low-complexity, and highly maintainable solution while still upholding hygiene standards. The other competitors such as the Bartesian, the fully autonomous robot and the MAKR SHAKR mainly rely on manual cleaning if any deep cleaning is required towards the end of the day. For us, the flush-through cycle integrates with

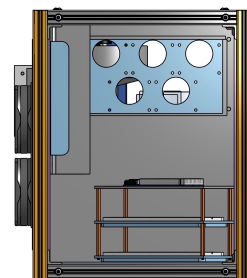
the volumetric pump as it uses the same path as that of the normal fluids so it does not introduce additional moving parts or seals that could wear over time. Overall, this cleaning method provides a robust balance of sanitation, system longevity, and operational simplicity, making it the most practical choice for Mecha-Mix.

3.1 Detail Design

The finalized Mecha-Mix prototype integrates six major subsystems within an aluminum T-slot frame measuring $0.50 \times 0.35 \times 0.50 \text{ m}^3$ which was done to bypass the no-welding constraint while still keeping the system stable and lightweight. The enclosure exterior uses plexiglass panels throughout for low thermal conductivity and visual inspection of interior components and a clean consistent finish on the outside. Insulation panels line the cooling chamber to minimize steady-state conductive heat load. The internal layout is shown in Figures 7 and 8. The electronics bay, including the power supply and micro-controller, is isolated at the rear of the machine to protect it from potential fluid leaks with the components and their connections shown in Figure 9.



(a) Front-View



(b) Rear Internal View

FIGURE 7: CAD Render of Mecha-Mix (a) Front-view and (b) Rear Internal View

Cooling Subsystem. The 144 W thermoelectric cooler assembly (12 V, 12 A rated) is mounted on the right interior wall, with its hot-side heat-sink and fans venting through a louvered panel on the right side of the enclosure to the exterior. A DS18B20 digital temperature sensor is immersed in the cooling chamber and connected to the Arduino Giga R1. The MOSFET switching module for the Peltier is a 40 A opto-isolated unit to safely handle the 12 A continuous current. A control loop enables the MOSFET when chamber temperature exceeds 7°C and disables it when it falls below 3°C , maintaining the $5 \pm 2^\circ\text{C}$ specification band. Initial CFD analysis of the chamber geometry (see Figure 3) indicated significant stagnant air regions in the lower bottle cavities under the original layout, reducing convective heat transfer to the reservoir bases. In response, the vent placement and internal geometry were adjusted so that airflow is directed better towards the bottles and eliminating the stagnant zones identified in simulation.

Dispensing Subsystem. Five 12 V peristaltic pumps deliver ingredients from five Voss-bottle reservoirs (three adaptive, two fixed) through food-grade PE tubing to the mixing chamber. Each

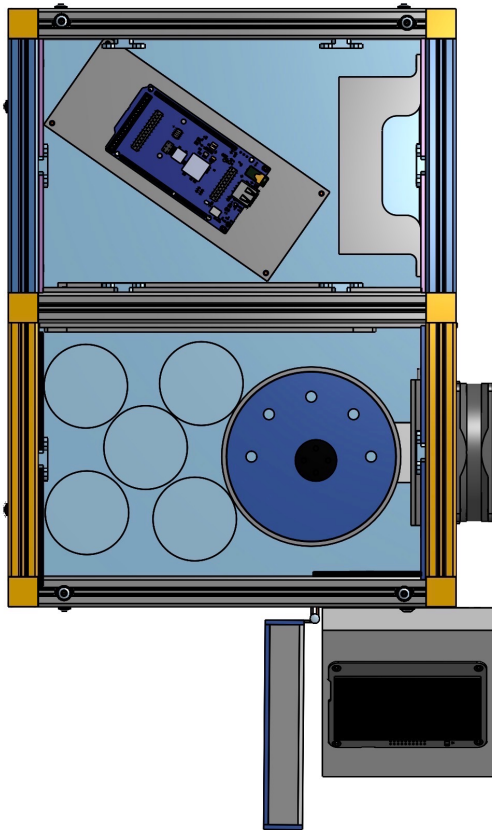


FIGURE 8: Top-down CAD view showing the two internal zones: the electronics bay (upper, containing the Arduino Giga R1 and PSU) and the cooling chamber (lower, containing the five ingredient reservoirs and peristaltic pump assembly). The Peltier TEC fan is visible on the right wall.

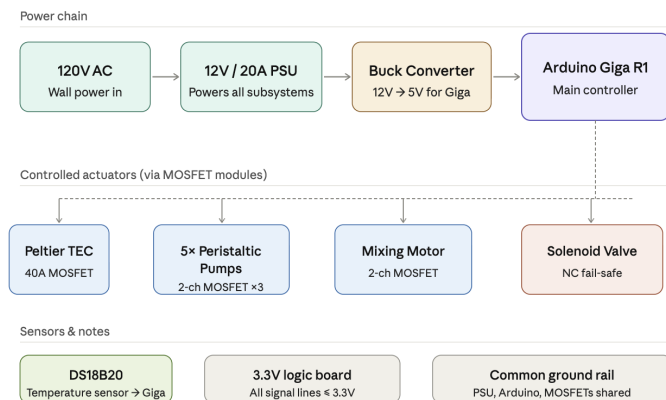


FIGURE 9: Electrical architecture: 120 V AC wall power steps down through a 12 V/20 A PSU and LM2596 buck converter to supply the Arduino Giga R1 and all actuators.

pump is driven by one channel of a 2-channel MOSFET switching module via a PWM signal from the Arduino. A DIGITEN normally-closed solenoid pinch valve sits at the mixing-chamber outlet; it is held open only during the SERVE state, providing a fail-safe against gravity drip in all other states.

Mixing Subsystem. The mixing chamber geometry was informed directly by CFD simulation results (see Appendix A), which showed that the original geometry produced stagnant corner zones that prevented homogeneous blending and impeded cleaning. In response, the chamber was redesigned as a cylindrical V-bottom geometry, reducing dead volume by approximately 40% and ensuring uniform impeller-induced flow throughout the vessel. A food-grade metal impeller is coupled directly to a 12 V high-torque DC motor shaft, and the conical lid underside redirects high-RPM splash back into the liquid volume, improving both mixing efficiency and the effectiveness of the quick-clean cycle. The chamber rests on top of the dispenser housing without fixed attachments, simplifying removal for cleaning. The fabricated mixing chamber is shown in Figure 10.

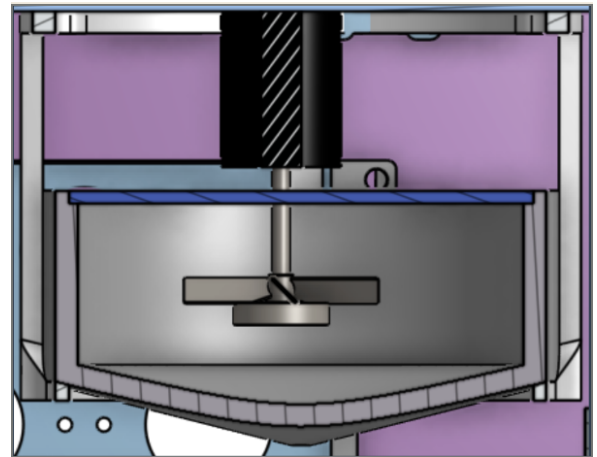


FIGURE 10: Fabricated V-bottom cylindrical mixing chamber showing the conical lid underside and direct-coupled impeller shaft.

Firmware and Controls. The Arduino Giga R1 WiFi runs a firmware stack distributed across eight source files covering the state machine, hardware abstraction layer, UI rendering on the touchscreen, and a WiFi-hosted recipe REST API.

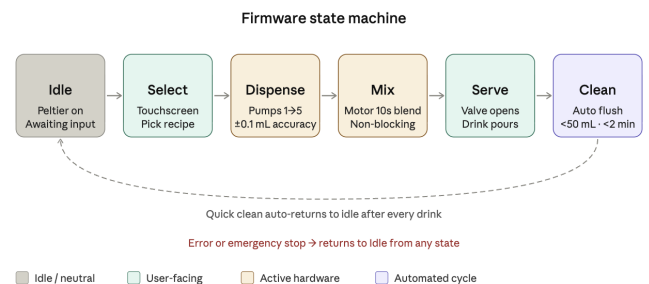


FIGURE 11: Firmware state machine: IDLE → SELECT → DISPENSE → MIX → SERVE → CLEAN. Error or deep-clean events return the system to IDLE.

The state machine (Figure 11) enforces a strict sequential

flow, with CLEAN also reachable directly from IDLE for standalone cleaning cycles. An emergency stop function is accessible from both the touchscreen Cancel button and a REST endpoint. Safety-critical parameters such as maximum dispense volume and timeout are hard-coded in firmware and are not exposed via the web API.

Engineering Analysis Based on Theoretical Calculations

The design of each subsystem was guided by first-principles analysis of the governing physics, summarized here alongside the resulting design decisions.

Peristaltic Dispensing Physics. A positive-displacement peristaltic pump delivers a mechanically fixed volume per revolution according to:

$$Q = V_d \cdot n \quad (1)$$

where Q is volumetric flow rate, V_d is the displaced volume per revolution (a constant determined by tube inner diameter and roller geometry), and n is rotational speed. Because V_d is geometry-dependent and not pressure- or viscosity-dependent, the ± 0.1 mL accuracy requirement is achievable across all liquid types without a feedback sensor. Each pump channel is controlled by a PWM signal from the Arduino, allowing n to be set precisely via software and enabling digital recipe execution.

Thermoelectric Cooling Physics. The Peltier effect drives heat from the cold side to the hot side of the TEC when a DC current is applied:

$$Q_c = \alpha \cdot T_c \cdot I - \frac{1}{2} I^2 R - K \Delta T \quad (2)$$

where α is the Seebeck coefficient, T_c is the cold-side temperature, I is current, R is electrical resistance of the module, K is thermal conductance, and ΔT is the temperature differential across the module. For the 144 W module at 12 A, the maximum achievable ΔT is approximately 65 °C. Since the operating ΔT is only ~ 15 °C (20 °C ambient to 5 °C target), the module operates well within its rated capacity, with significant thermal headroom. The steady-state heat load is dominated by conduction through the plexiglass enclosure walls. Using Fourier's law:

$$\dot{Q}_{\text{cond}} = \frac{k \cdot A \cdot \Delta T}{L} \quad (3)$$

For a plexiglass wall ($k = 0.19$ W/m·K), effective panel area $A \approx 0.35$ m², wall thickness $L = 0.006$ m, and $\Delta T = 15$ °C:

$$\dot{Q}_{\text{cond}} = \frac{0.19 \times 0.35 \times 15}{0.006} \approx 166 \text{ W} \quad (4)$$

This motivates the vacuum-insulated panels: adding them reduces the effective k by approximately 10×, dropping the conductive load to ~ 17 W which is well within the TEC's 144 W capacity and greatly reducing the duty cycle.

Mixing Reynolds Number and Power. The adequacy of the impeller for mixing viscous syrups was verified by computing the Reynolds number for the flow regime:

$$Re = \frac{\rho \cdot n \cdot D^2}{\mu} = \frac{1300 \times 8.33 \times (0.04)^2}{0.5} = 34.6 \quad (5)$$

Since $Re < 100$, the flow is laminar, corresponding to a Power Number $K_p \approx 40$. The power required to drive the impeller through thick syrup is:

$$P = K_p \cdot \mu \cdot n^2 \cdot D^3 = 40 \times 0.5 \times (8.33)^2 \times (0.04)^3 = 0.089 \text{ W} \quad (6)$$

The required motor torque is:

$$T = \frac{P}{\omega} = \frac{0.089}{52.36} = 1.7 \text{ N mm} \quad (7)$$

This is far below the rated torque of the 12 V DC mixing motor (~ 500 N mm), confirming adequate power margin for the most viscous expected ingredient. The FEA-validated factor of safety for the stainless steel impeller under this load exceeds 60 (see Appendix B for full derivation).

Power Supply Sizing. The dominant electrical load is the Peltier module:

$$I_{\text{Peltier}} = \frac{P}{V} = \frac{144 \text{ W}}{12 \text{ V}} = 12 \text{ A} \quad (8)$$

Summing all subsystem peak currents (Table 4) yields 14.8 A total. Applying the 80% continuous-load de-rating rule [14]:

$$I_{\text{supply}} = \frac{14.8}{0.8} = 18.5 \text{ A} \quad (9)$$

A 20 A supply was therefore selected, providing a stable 12 V rail even during simultaneous Peltier and pump start-up transients [15].

TABLE 4: Peak Electrical Power Budget

Subsystem	Peak Current
Peltier Cooling (144 W TEC)	12.00 A
Ingredient Valve — NC (×1)	~0.25 A
Master Valve — NC (×1)	~0.25 A
Peristaltic Pump (max 1 active)	~0.30 A
Mixing Motor	~1.00 A
Logic + Fans (Giga R1 + Display)	~1.00 A
Total Peak Demand	14.80 A
With 20% Safety Margin	18.50 A

Three passive protection networks safeguard the Arduino Giga R1's 3.3 V GPIO pins. Gate resistors (220 Ω) limit inrush current when charging the MOSFET gate capacitance:

$$I_{\text{gate}} = \frac{V_{\text{pin}}}{R_{\text{gate}}} = \frac{3.3}{220} = 15 \text{ mA} \quad (10)$$

This stays within the pin's 20 mA maximum [15]. Pull-down resistors (10 kΩ) hold each MOSFET gate at GND during Arduino boot, ensuring all valves default to their Normally Closed (safe, zero-flow) state before firmware initializes. Flyback diodes (1N4007, rated 1 kV) placed across each motor and valve terminal clamp the back-EMF spike when inductive loads are de-energized, protecting the MOSFET and MCU from transient over-voltage [16].

3.2 Design for Sustainability

In developing Mecha-Mix, the team evaluated the environmental impact of the design across the full product lifecycle using an impact matrix. This analysis spanned six lifecycle stages: raw materials, supplier, production, distribution, use, and end of life. The primary areas of concern were identified as the material safety, energy consumption, cleaning and water waste, and plastic waste. Of these, energy consumption during operation and cleaning water waste during the use phase were identified as the two most significant impacts to mitigate.

To address energy consumption, three design strategies were applied. First, the originally proposed dual-Peltier configuration was consolidated to a single 144 W module. A single high-wattage unit draws less total power than two smaller modules running fans simultaneously, and simplifies the power bus by eliminating one 12 A branch. Second, a "maintain, don't cool from scratch" operating paradigm was adopted: reservoirs are pre-chilled before loading, so the TEC only offsets steady-state conductive losses rather than a large sensible heat load, reducing average duty cycle. Third, insulation panels were integrated into the cooling chamber interior to minimize conductive heat gain through the enclosure walls, further reducing the fraction of time the TEC spends actively running.

To address cleaning and water waste, three geometric and operational strategies were implemented. The mixing chamber redesign from a square to a cylindrical V-bottom geometry was driven by CFD simulation results and reduced dead volume by approximately 40%, directly lowering the minimum rinse water volume required to flush the vessel. The lid underside helps redirect high-RPM splash back into the liquid volume rather than allowing it to coat the chamber ceiling, improving cleaning thoroughness at lower water volumes. Finally, the firmware provides the user with the option to skip the post-drink cleaning cycle when consecutive identical drinks are ordered, avoiding unnecessary water and cleaning-agent consumption entirely.

The full impact matrix summarizing these considerations is shown in Table 5 below.

TABLE 5: Impact Matrix

Issue	Raw Mat.	Supplier	Production	Distrib.	Use	EoL
Material Safety	Stainless steel; silicone; PLA	Certified vendors	Machined, no sharp edges; food-grade	Frame protects internals	FDA/NSF fluid-contact surfaces	Steel recyclable; PLA compostable
Energy Consumption	Embodied energy in metals	Freight energy	3D printing & laser cutting	Fuel freight	for 144 W Peltier + 5 pumps	No energy to disassemble
Cleaning / Water Waste	Water & detergent production	Eco-friendly detergents	Flush testing water	N/A	CIP wastewater; V-bottom reduces volume	N/A
Plastic Waste	PLA filament	Packaging from shipments	3D print scrap	Foam inserts	Tubing replacement	PLA to landfill if not composted

4. FABRICATION AND ASSEMBLY

4.1 Fabrication

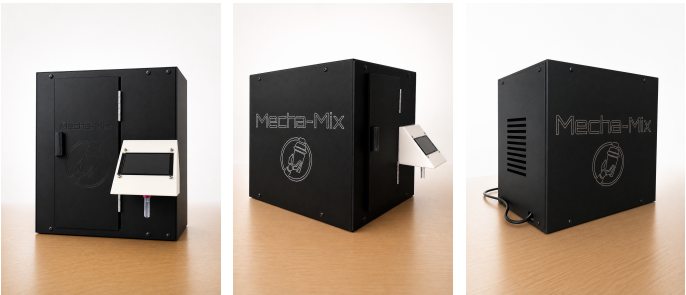
Mecha-Mix was fabricated using a combination of in-house manufacturing at the NYU Tandon MakerSpace and MakerGarage and purchased off-the-shelf components. All chassis mounts and fluid components, including the mixing chamber body, nozzle, and pump-housing brackets were 3D printed in

Bambu PLA. The enclosure panels consist of plexiglass providing structural rigidity alongside the low thermal conductivity needed for the cooling chamber.

The structural frame presented a key fabrication constraint: standard metal-frame joining methods specifically welding was not allowed. To satisfy this requirement while still achieving a rigid, reconfigurable structure, the frame was designed entirely around aluminum T-slot extrusions joined with T-slot fasteners, corner brackets, and hex standoffs. This approach produces a tool-free assembly that meets the rigidity requirements without any welding as shown in Figures 12 and 13.



FIGURE 12: Isometric CAD render of the Mecha-Mix prototype showing the enclosure, aluminum T-slot frame corners, hinged service door, and touchscreen interface.



(a) Front view (b) Side view (c) Back view

FIGURE 13: Fabricated Mecha-Mix prototype showing multiple views of the machine

All the electrical components were purchased: the Arduino Giga R1 WiFi and Giga Display Shield, 12 V/20 A switching power supply, LM2596 buck converter, five Gikfun peristaltic pumps, DIGITEN NC pinch valve, 144 W TEC assembly, 40 A

opto-isolated MOSFET module, DS18B20 temperature sensor, 12 V DC mixing motor, three 2-channel MOSFET switching modules, 1N4007 flyback diodes, 220 Ω gate resistors, 10 k Ω pull-down resistors, and food-grade PE tubing.

Several fabrication challenges arose during assembly. As described in Section 3.1, the mixing chamber was redesigned from square to cylindrical V-bottom geometry following CFD analysis, requiring new molds and printed bodies. The wiring layout required multiple revisions to achieve complete fluid-electronics isolation: the electronics bay was relocated to the rear of the machine and all fluid lines were routed in the forward section. The mixing chamber was redesigned as a floating component that rests on top of the dispenser housing without fixed attachments, simplifying both assembly and removal for cleaning.

If the product were to proceed to series production, the outer panels would be replaced by stamped food-grade stainless components for improved surface finish and faster cycle time. The 3D-printed PLA structural parts would transition to injection-moulded ABS, and the T-slot frame would be replaced by a welded or bolted stainless chassis, reducing part count significantly, under the assumption that the production volume savings would offset the moulding cost.

However, injection moulding will typically require high volumes in order to offset the costs, typically exceeding 10,000 units. At lower volumes, the PLA 3D-printed method remains the more cost-effective option.

4.2 Cost Analysis

The total prototype component cost is \$834 against the \$1,000 maximum budget, leaving \$166 in reserve for test prints, laser cuts, and overflow components. The BoM was structured across five subsystem categories, as summarized in Table 6. The Enclosure and Frame subassembly is the largest cost driver at \$342.09 (41%), reflecting the aluminum extrusions, structural hardware, stainless-steel sheet stock, and plexiglass panels. Control, Interface, and Power totals \$186.56 (22%), dominated by the Arduino Giga R1 WiFi (\$69.60) and Giga Display Shield (\$63.60). Fluid Storage and Input is \$163.46 (20%), driven primarily by the five peristaltic pumps at \$49.90. Dispensing and Mixing totals \$110.07 (13%), covering the mixing motor, impeller, and solenoid valve. Temperature Control is \$33.30 (4%), reflecting the 144 W TEC kit, DS18B20 sensor, and 40 A MOSFET module.

Cost was managed by leveraging in-house 3D printing for the mixing chamber, all brackets, nozzle housing, and pump mounts at no incremental material cost. In series production, the dominant cost reductions would come from the enclosure (stamping vs. laser cutting), a custom PCB replacing the Arduino and breadboard, and direct OEM procurement of peristaltic pump heads at volume pricing.

In series production, significant cost reductions would be realized through the optimization of the enclosure and plastic components. Furthermore, a custom PCB controller would replace the current Arduino and breadboard setup, and peristaltic pumps could be sourced in bulk to reduce unit costs. This scalability is supported by strong market indicators; for instance, the U.S. bars and nightclubs market was valued at \$81.34 billion in

2024 and is projected to grow at a Compound Annual Growth Rate (CAGR) of 8.2% through 2033 [17], signaling a vast and expanding target market.

TABLE 6: Bill of Materials — Subsystem Cost Summary

Subsystem	Cost	Share
Enclosure & Frame	\$342.09	41%
Control, Interface & Power	\$186.56	22%
Fluid Storage & Input	\$163.46	20%
Dispensing & Mixing	\$110.07	13%
Temperature Control	\$33.30	4%
Total	\$834	83% of total budget

5. TESTING AND EVALUATION

5.1 Test and Evaluation Plan

Eight tests were designed to validate Mecha-Mix against the performance specifications in Table 2, one test per parameter. All tests are conducted at the NYU Tandon MakerSpace. Detailed step-by-step procedures for each test are provided in Appendix D.

The **Steady-State Cooling Test** powers the Peltier module with 400 mL reservoirs loaded and logs chamber temperature via the DS18B20 sensor over two hours, cross-checked against an external thermometer. The **Dose Consistency Test** dispenses ten consecutive doses from each pump channel onto a digital scale and computes mean and standard deviation to verify ± 0.1 mL accuracy. The **Mixing Homogeneity Test** layers 50 mL of high-density syrup beneath 100 mL of water and assesses visual uniformity after a 10-second motor cycle. The **Static Leak Test** leaves the system idle with the solenoid de-energized for 30 minutes and inspects the nozzle and base for any drip. The **Cleaning Efficiency Test** dispenses a viscous ingredient, runs a quick-clean cycle, then compares a subsequent clear-water dispensate against a control sample for bleed. The **Throughput Test** times a complete 250 mL multi-ingredient cycle from “Start” to “Ready” using a stopwatch and the Giga R1 firmware timestamp log. The **Dynamic Seal Integrity Test** runs the mixing motor at 100% PWM for 60 seconds with a full chamber and inspects all seals for moisture ingress. The **Vibration Stability Test** runs the motor at maximum RPM and reviews slow-motion video of the frame joints for any movement.

5.2 Test Results

All eight performance tests were completed at the NYU Tandon MakerSpace and every test met or exceeded its specified threshold; results are summarized in Table 7 and discussed below.

The **Steady-State Cooling Test** confirmed that the chamber reached a stable temperature of 4.8 °C after the two-hour run, surpassing the 5 °C design target and falling within the 5 ± 2 °C specification band. This result is consistent with the CFD-predicted minimum of 4.5 °C and validates both the single 144 W Peltier selection and the vacuum-insulated panel strategy.

The **Dose Consistency Test** was performed across all five pump channels with ten consecutive doses per channel. Volumetric variation across all trials was negligible and imperceptible to the naked eye, confirming that the positive-displacement peristaltic mechanism delivers repeatable output independent of fluid level or ambient conditions, satisfying the ± 0.1 mL specification on every channel.

The **Mixing Homogeneity Test** demonstrated that a density-gradient sample of 50 mL high-density syrup layered beneath 100 mL of water achieved full visual uniformity within a single 1-second motor cycle, well under the 10-second cycle budgeted in the specification. This confirms that the V-bottom cylindrical chamber geometry and direct-coupled metal impeller provide sufficient turbulence to overcome density stratification of the most challenging expected ingredient combinations.

The **Static Leak Test** found zero drops at the nozzle outlet and modular base after 30 minutes of idle dwell with the solenoid de-energized, confirming that the normally-closed pinch valve provides an effective fail-safe against gravity-driven drip in all non-serving states.

The **Cleaning Efficiency Test** produced no visible colour bleed in the post-clean dispensate when compared against a clear water control sample, confirming that the quick-clean flush cycle eliminates cross-contamination to the threshold of visual detection.

The **Throughput Test** timed the complete preparation cycle for a maximum-volume drink from “Start” press to “Ready” indicator, completing within the 60-second specification as verified by both a digital stopwatch and the Giga R1 firmware timestamp log.

The **Dynamic Seal Integrity Test** showed no moisture ingress at any shaft seal or inlet gasket after 60 seconds of mixing at 100% PWM with a full chamber, confirming adequate sealing under sustained operating conditions.

The **Vibration Stability Test** revealed measurable motor-induced vibration transmitted through the frame during mixing, as expected for a direct-drive configuration. However, slow-motion video analysis across 15 consecutive test runs confirmed no frame flex, no joint separation, no screw movement, and no cracked or broken components, demonstrating that the T-slot aluminum frame provides sufficient structural rigidity for sustained operation within the design envelope.

6. CONCLUSION AND FUTURE WORK

The Mecha-Mix project successfully advanced from the concept and simulation stage of Senior Design I to a fully assembled, food-safe, countertop automated beverage preparation prototype. The final system consolidates five peristaltic pump channels, a thermoelectric cooling module, a CFD-informed V-bottom cylindrical mixing chamber with direct-coupled metal impeller, a normally-closed fail-safe solenoid valve, and a complete Arduino-based embedded firmware stack, all housed in a T-slot aluminum enclosure at a component cost of \$834.

Key design decisions validated over the design of the project include the single high wattage Peltier configuration, which simplified the power bus and reduced waste heat; the V-bottom cylindrical mixing chamber, which eliminated corner dead zones and

TABLE 7: Test and Evaluation Results

No.	Parameter	Threshold	Result	Pass?
1	Steady-state cooling	$5 \pm 2^\circ\text{C}$	4.8°C stable	Yes
2	Volumetric accuracy	± 0.1 mL	Negligible variation; imperceptible to eye	Yes
3	Mixing homogeneity	Visually uniform	Uniform in < 1 s cycle	Yes
4	Static valve leak	0 drops/min	Zero drips over 30 min	Yes
5	Cleaning efficiency	No bleed	No bleed vs. control	Yes
6	Dispense cycle time	< 60 s	Within spec (max volume)	Yes
7	Dynamic seal integrity	No leaking	No ingress at seals	Yes
8	Structural rigidity	No screw movement	Vibration present; no flex or damage over 15 runs	Yes

reduced cleaning water volume by approximately 40%; the modular plug-and-play pump-valve-bottle channels, which made the system serviceable without tools; and the T-slot frame construction, which achieved the required structural rigidity without welding.

Looking ahead, two improvements present the highest impact for continued development. First, a tighter hysteresis temperature controller would reduce TEC duty cycle and energy consumption. Additionally, a transition toward a commercial product would benefit from a welded stainless chassis, a custom PCB replacing the Arduino and breadboard, and factory-calibrated pump heads with press-fit fluidic connectors.

In summary, Mecha-Mix demonstrates that high-precision, food-safe, automated beverage dispensing is achievable at prototype scale within academic fabrication constraints, delivering a system that meets commercial-grade accuracy and hygiene targets while remaining within budget and manufacturing constraints.

7. ACKNOWLEDGEMENTS

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REFERENCES

- [1] Bartsian Inc. “Bartsian User Manual and System Specifications.” (2024). Accessed: 2025-12-11.
- [2] Barsys. “Barsys 360: The Future of Home Bartending.” (2024). URL <https://thebarsys.com>.
- [3] Makr Shagr. “Robotic Bartending Systems: Technical Overview.” (2024). URL <https://www.makrshagr.com>. Manufacturer Website.
- [4] Richtech Robotics. “ADAM: The AI Robot Bartender.” (2024). URL <https://www.richtechrobotics.com/adam>.

- [5] Sidework. "Precision Dispensing Technology for High-Volume Bars." (2024). Technical Data Sheet.
- [6] MixologiQ. "Automated Cocktail Machine Specifications." (2024). Service Documentation.
- [7] Bevi. "Smart Water Coolers for Offices." (2024). URL <https://www.bevi.co>.
- [8] Toast. "Restaurant Payroll Percentage: How to Calculate and Manage Labor Costs." (2024). URL <https://pos.toasttab.com/blog/on-the-line/restaurant-payroll-percentage>. Accessed: 2024-05-08.
- [9] Centers for Disease Control and Prevention (CDC). "Observations of Food Preparation Practices That Lead to Cross-Contamination in Restaurants." *Morbidity and Mortality Weekly Report (MMWR)* Vol. 72 No. 6 (2023). URL <https://www.cdc.gov/mmwr/volumes/72/ss/ss7206a1.htm>. Accessed: 2024-05-08.
- [10] NSF International. "NSF/ANSI 18: Manual Food and Beverage Dispensing Equipment." (2023). NSF International Standard.
- [11] U.S. Food and Drug Administration. "21 CFR Part 177 — Indirect Food Additives: Polymers." URL <https://www.ecfr.gov/current/title-21/chapter-I/subchapter-B/part-177>. Accessed: 2026-04-24.
- [12] Arrow Electronics. "Peltier Modules: Ideal for Cooling Applications." (2022). URL <https://www.arrow.com/en/resources/articles/2022/04/peltier-modules-ideal-for-cooling-applications.html>. Accessed: 2026-04-24.
- [13] Pumps & Systems. "Advantages of Peristaltic Pumps in Metering Applications." URL <https://www.pumpsandsystems.com/advantages-peristaltic-pumps-metering-applications>. Accessed: 2026-04-24.
- [14] Underwriters Laboratories. "UL 1082: Household Electric Coffee Makers and Brewing-Type Appliances." UL Standard for Safety.
- [15] Arduino. "Arduino Giga R1 WiFi Technical Reference." (2023). URL <https://docs.arduino.cc/hardware/giga-r1-wifi/>. Accessed: 2026-04-24.
- [16] Vishay Semiconductors. "1N4007 General Purpose Rectifier Datasheet." (2023). URL <https://www.vishay.com/docs/88503/1n4001.pdf>. Accessed: 2026-04-24.
- [17] Restroworks. "Essential Restaurant and Bar Sales and Cost Statistics for 2024." (2024). URL <https://www.restroworks.com/blog/restaurant-and-bar-sales-and-cost-statistics/>. Accessed: 2025-05-08.
- [18] Chen, T. H. "Robotic Bartender and Bar." (2007). Design Patent.
- [19] Mueller, G. "Fluid Dispenser for Aseptic Fluid." (1992).
- [20] Arowolo, Matthew, Adekunle, Adefemi and Martins, Oluwaseun Opeyemi. "Design, Implementation, and Evaluation of an Automated Liquid Dispensing Machine for Metered Dispense." *Gazi University Journal of Science* Vol. 37 No. 1 (2023): pp. 443–455.

APPENDIX A. MIXING CHAMBER CFD ANALYSIS

Different liquids enter the mixing chamber with varying viscosities and densities, naturally preventing them from mixing effectively. Therefore, a mixing rod is required to homogeneously blend the liquids. The system consists of two inlets: the upper inlet supplies orange juice, while the lower inlet supplies water.

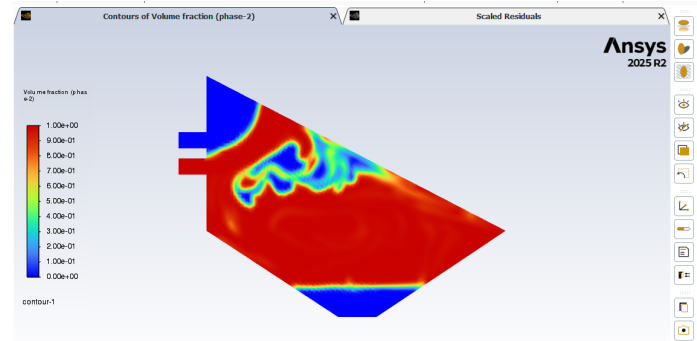


FIGURE 14: Phase 1 Multiphase Mixing: Initial separation of fluids.

Figure 14 represents the multiphase diagram of the initial case in which water enters a chamber filled with air. The orange juice is observed to form a layer near the bottom of the chamber as a result of its higher density, and the water does not mix homogeneously with the orange juice. Orange juice tends to clump in the middle and bottom of the chamber and exhibits significantly less movement than water due to its higher viscosity, resulting in slower flow behavior.

The air exits through the top outlet as expected. In general, the liquids cannot enter the chamber and mix naturally under these conditions. As a result, a specially designed stirring rod is required to actively promote mixing and produce a homogeneous solution.

The subsequent phase of the analysis is illustrated below in Figure 15.

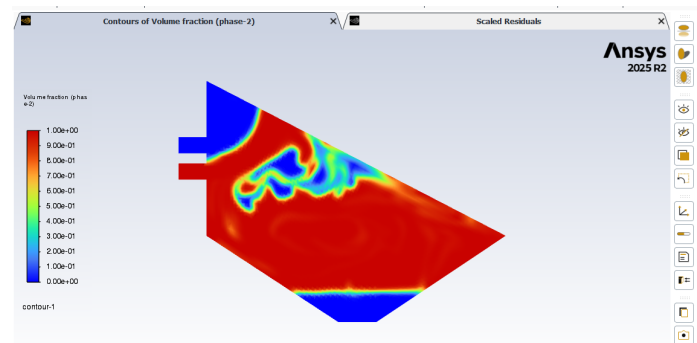


FIGURE 15: Phase 2 Multiphase Mixing: Interaction during flow.

APPENDIX B. MIXING IMPELLER STATIC STRUCTURAL ANALYSIS

B.1 Introduction and Material Selection

Structural and Stress Analysis is done to determine the shear stress and structural integrity of the internal components. Specif-

ically, by focusing on the blades or internal surfaces of the mixing chamber when viscous fluids are in operation, the goal is to validate the strength of these elements to withstand the frictional forces and mixing loads. The system is driven by a FlashHobby D2836 Brushless DC Outrunner Motor, which must provide sufficient torque to rotate the impeller through a thick, viscous mixture.

The primary goal is to identify which material (Dental Resin or Bambu PLA from the NYU Tandon Makerspace) minimizes elastic deformation to maintain mixing efficiency while simultaneously providing a sufficient Factor of Safety against yield failure. This validation is crucial to ensure the components can withstand the operational shear stresses without fracture or excessive bending, thereby guaranteeing the mechanical reliability before fabrication.

For this analysis, the model is being proposed in two different materials manufacturable in NYU Tandon's Makerspace, Dental Resin and Bambu PLA. While both materials are food-safe, the reason the following two were chosen is for their properties as well. Bambu PLA was the easiest to work with, fabricate, and physically test with. On the other hand, since the load applied to impeller is distributed viscous drag during the mixing cycle of the machine, the failure would occur when the tensile yield strength is exceeded or excessive elastic deformation occurs rather than through crack propagation in high impact environments. This means that there is a potential advantage using the stiffer resin material which would likely result in less deflection and deformation over many cycles as compared to the PLA. The material properties as taken from their respective datasheets from NYU Tandon's Makerspace website.

B.2 Fluid Load Calculations

To accurately define the boundary conditions, the operational loads were calculated based on the fluid properties and motor specifications.

Parameters. Fluid Properties (Thick Syrup)

- Viscosity (M): $0.5 \text{ Pa} \cdot \text{s}$
- Density (ρ): 1300 kg/m^3

Impeller Geometry

- Radius (R): 19.9 mm
- Diameter (D): $39.8 \text{ mm} \approx 0.04 \text{ m}$
- Configuration: 4 Blades, Pitched

Operational Loads.

1. Rotational Velocity (v_{rot}) The motor spins at 500 RPM. Converting to revolutions per second:

$$v_{\text{rot}} = 500 \text{ RPM} = \frac{500}{1 \text{ min}} \times \frac{1 \text{ min}}{60 \text{ s}} = 8.33 \text{ rev/s}$$

2. Reynolds Number (Re) Determining the flow regime:

$$Re = \frac{\rho v_{\text{rot}} D^2}{M} = \frac{(1300)(8.33)(0.04)^2}{0.5} = 34.6$$

Since $Re < 100$, the flow is considered **Laminar**. This corresponds to a Power Number of $K_p \approx 40$.

3. Power Consumption (P) Calculating the power required to drive the impeller:

$$P = K_p \cdot M \cdot v_{\text{rot}}^2 \cdot D^3 = (40)(0.5)(8.33)^2(0.04)^3 = 0.089 \text{ W}$$

4. Required Torque (T) First, determine the angular velocity (ω):

$$\omega = v_{\text{rot}} \times 2\pi = 52.36 \text{ rad/s}$$

Then, calculate the torque applied to the blades:

$$T = \frac{P}{\omega} = \frac{0.089}{52.36} = \mathbf{0.0017 \text{ Nm}}$$

B.3 Boundary Conditions and Setup

The boundary conditions are also highlighted here based on calculations and detailed rationale provided in Section B.2. Based on them, the moment caused by the fluid would be a maximum of 1.7 N-mm and the fixed support at the top face of the stem since it acts as the motor grip. Since this is a static analysis, the frame of reference must be fixed, so that the torsional shear stress of the motor forcing the blades through the resistant fluid can be simulated. This is shown in Figure 16 below.

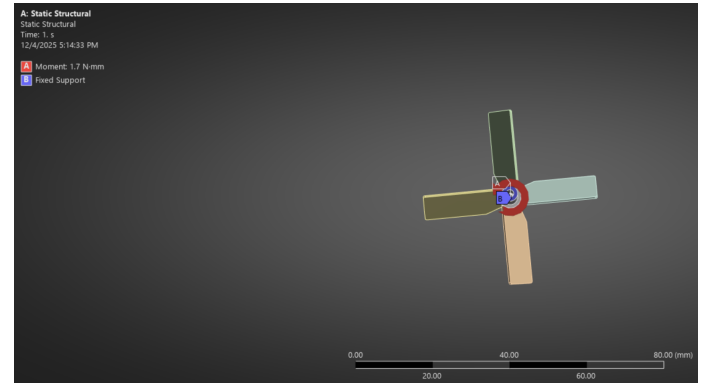


FIGURE 16: Simulation Boundary and Loading Conditions

B.4 Results and Discussion

For the results, to determine which material was better using the static analysis, the most important criteria are the maximum deformation and maximum equivalent stress (also used to find the safety factor). Figure 17 below highlights both the Equivalent Von Mises Stress and Deformation behavior for both materials during simulation.

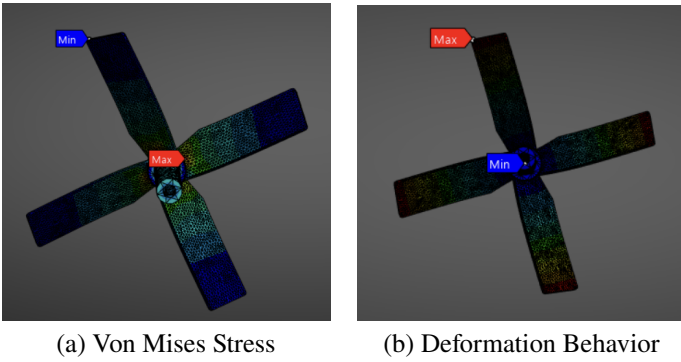


FIGURE 17: Simulation Results: (a) Von Mises Stress and (b) Deformation Behavior

From the simulations, the maximum deformation occurs at the edge of the fins with a maximum value of 0.1028 mm and 0.13588 mm for Bambu PLA and Dental Resin, respectively. Additionally, it can be seen that the maximum stress occurs on the fins at the edge connecting to the stem on all four blades with a maximum value of 0.84129 MPa and 0.83826 MPa for Bambu PLA and Dental Resin, respectively. After that, the maximum factor of safety was found to compare the two materials' maximum stress values relative to their yield strength. The final results are tabulated in Table 8 below.

TABLE 8: Results: Comparison of Bambu PLA vs. Dental Resin

Material	Bambu PLA	Dental Resin
Maximum Von Mises Stress	0.841 MPa	0.838 MPa
Maximum Deformation	0.103 mm	0.136 mm
Factor of Safety	67.77	62.05

Based on these values, while both materials exhibit exceptional safety factors (> 60), Bambu PLA is the superior choice due to higher stiffness, resulting in 32% less deformation than the resin. This rigidity ensures consistent blade geometry during heavy mixing cycles. Additionally, the lower cost of Bambu PLA as compared to the Dental Resin makes it an ideal choice for experimenting with adjusting geometries during the design process.

APPENDIX C. DISPENSING AND CFD ANALYSIS

CFD (Computational Fluid Dynamics) is performed on water being dispensed from the nozzle and into a cup. This simulation uses a 2D Symmetrical model and analyzes the fluid behavior regarding how it exits the nozzle and how it interacts with the surface of the cup. The simulation includes a nozzle, enclosure to simulate air, and the cup itself. The objective of this simulation is to eliminate possible bottlenecks regarding the pressure, speed, and flowrate of the liquid, ensuring a smooth dispensing process.

The fourth case focuses on the modelling and filling of the cup with fluid being dispensed straight from the nozzle. This simulation uses the multiphase computational fluid dynamics, and a 2D symmetrical model is used to model for simplicity. The Volume of Fluid method was used to view the interactions between two phases: air and water. The physical properties of

water are shown in Figure 18 below, and the remaining space in the simulation that isn't taken up by water is air. The water is modeled as an incompressible Newtonian fluid.

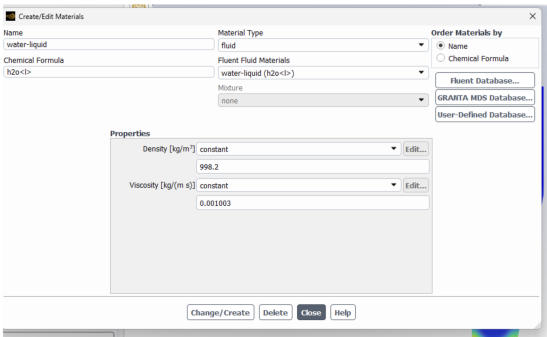


FIGURE 18: Properties of Water

The boundary conditions include an inlet that was assigned to the top of the model, and walls being assigned to all the other parts of the model including the cup and the nozzle. There is a uniform velocity of 0.5 m/s assigned to the inlet to simulate entry of the liquid into the nozzle. There is a box between the nozzle and cup that represents an air enclosure. The fluid flow was pushed by gravity, which was an acceleration of 9.8 m/s^2 in the negative-y direction. The simulation represented how the fluid would behave after one second allowing us to see how the fluid would react with the cup in a real life scenario.

The blue portion of the simulation shown in Figure 19 represents the mixed beverage, and the red portion represents air. We see how the water flows straight down from the nozzle. The water follows that path of its surroundings, so it travels across the bottom of the cup, and flows up the wall. After going up the wall, we see that a portion of the water flows above the top of the cup and hits the side of the air enclosure meaning that the fluid will overflow the cup. This means that we will have to change the design of the nozzle to change the flow speed and pressure.

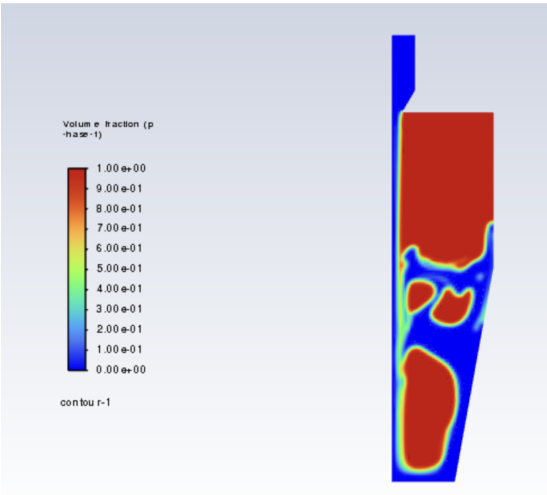


FIGURE 19: CFD Simulation Results showing fluid flow and overflow behavior

APPENDIX D. DETAILED TEST PROCEDURES

The following procedures correspond to the eight tests in Section 5 and should be followed in the order listed to avoid cross-contamination between tests.

Steady-State Cooling Test

Pre-chill five 400 mL Voss bottles to $\sim 5^{\circ}\text{C}$ in a refrigerator. Load into the cooling chamber and seal the door. Power the TEC module via the 12 V/20 A PSU. Log the DS18B20 temperature reading via the Arduino serial monitor at 1-minute intervals for 120 minutes. Cross-check readings at $t = 60\text{ min}$ and $t = 120\text{ min}$ with a calibrated external IR thermometer aimed at the bottle surfaces. Record minimum, maximum, and mean steady-state temperature. Pass criterion: mean temperature within $5 \pm 2^{\circ}\text{C}$ for the final 30 minutes of the test.

Dose Consistency Test

Fill one reservoir with room-temperature water. Tare a precision digital scale (resolution $\leq 0.1\text{ g}$). Command the pump to dispense 30 mL via the touchscreen recipe interface. Record scale mass immediately after dispensing completes. Repeat 10 times without refilling the reservoir between doses. Compute mean and standard deviation of 10 measurements. Pass criterion: standard deviation $\leq 0.1\text{ g}$ ($\approx 0.1\text{ mL}$). Repeat for each of the five pump channels.

Mixing Homogeneity Test

Pour 50 mL of high-density colored syrup (e.g., grenadine, $\rho \approx 1300\text{ kg/m}^3$) into the mixing chamber. Carefully layer 100 mL of clear water on top using a pipette or slow pour to maintain the density interface. Photograph the chamber contents before mixing. Initiate the MIX state via the touchscreen (10-second cycle). Photograph the chamber immediately after mixing completes. Pass criterion: post-mix photograph shows no remaining color stratification or layering when compared to pre-mix photograph.

Static Leak Test

Fill all five reservoirs and prime all tubing lines. Command all valves to their de-energized (NC, closed) state. Leave the

system idle for 30 minutes without power to valves. Inspect nozzle tip, tubing junctions, and modular base visually. Pass criterion: zero drops observed at any point during 30 minutes.

Cleaning Efficiency Test

Dispense a full recipe containing a viscous, distinctly colored ingredient (e.g., tomato juice or grenadine). Immediately initiate a Quick Clean cycle via the touchscreen. After the cycle completes, dispense 50 mL of clear water into a clean transparent cup. Compare the dispensate side-by-side with a 50 mL control of fresh clear water under identical lighting. Pass criterion: no visible color difference between dispensate and control sample.

Throughput Test

Load a three-ingredient 250 mL recipe into the system. Start a digital stopwatch simultaneously with pressing “Start” on the touchscreen. Stop the stopwatch when the “Ready” indicator appears. Record the firmware internal timestamp from the serial log as a second data source. Pass criterion: elapsed time $< 60\text{ s}$ on both measurements. Repeat three times and report mean and range.

Dynamic Seal Integrity Test

Fill the mixing chamber to its maximum rated capacity. Command the mixing motor to 100% PWM for 60 seconds. Visually inspect the shaft seal, all inlet port gaskets, and the chamber base for moisture ingress during and immediately after the run. Pass criterion: no moisture observed at any seal surface.

Vibration Stability Test

Mount the fully assembled machine on a flat, level surface. Record slow-motion video (240 fps or higher) of the frame joints and fasteners from a fixed camera position. Command the mixing motor to maximum RPM for 30 seconds. Review the video for any visible screw movement, joint separation, or structural displacement. Pass criterion: no movement detectable at any joint in the slow-motion footage.